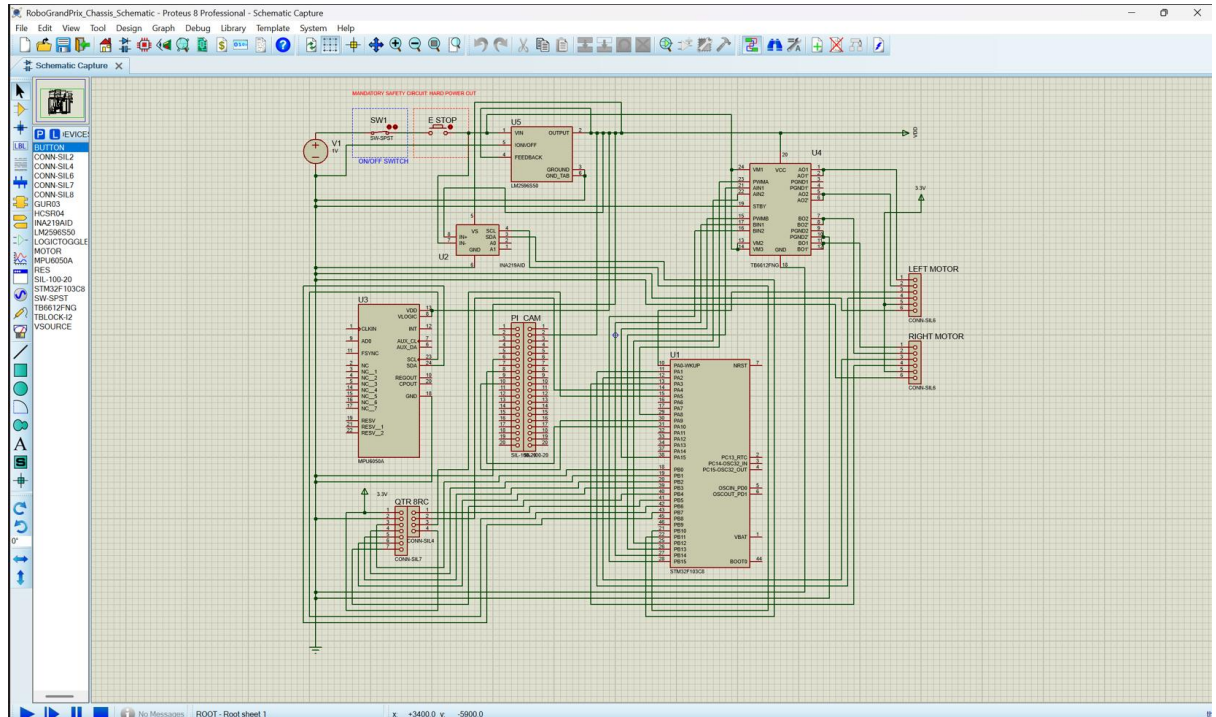


Electronic Design and Component Justification

Circuit Schematics & E-Stop Integration

The electronic architecture relies on an STM32 Blue Pill microcontroller interfacing with a TB6612FNG motor driver, stepping down power via a custom distribution board.

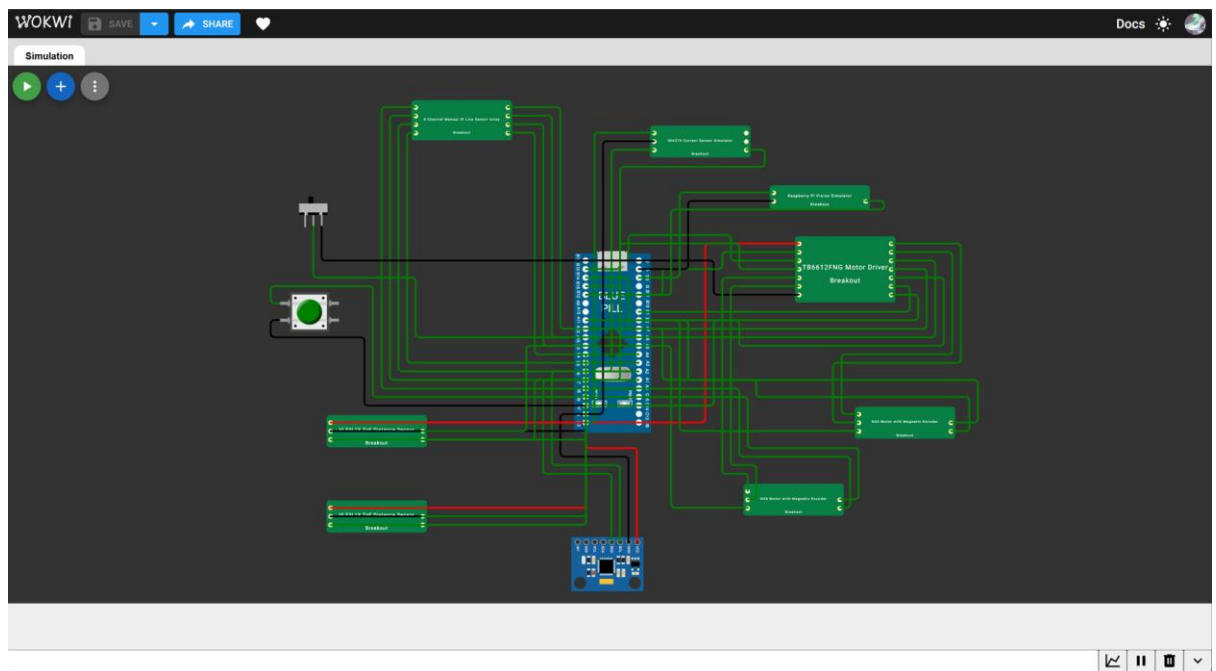
- **Proteus Schematic:**



This diagram details the physical routing of the PCB. **CRITICAL SAFETY NOTE:** The mandatory Emergency Stop (E-Stop) button and the main ON/OFF latching switch are explicitly annotated here, demonstrating how they physically interrupt the motor driver power loop.

- **Raw Proteus Project File:** The original CAD project file (RoboGrandPrix_Chassis_Schematic.pdsprj) is included in the repository's Schematics folder for complete transparency and deep technical review of the PCB routing, layers, and component footprints.

- **Wokwi Logic Map:**



The I2C bus wiring (IMU, INA219, ToF sensors) and the encoder feedback loops used for our PID state-machine simulation.

Processing and Control Units

- **Primary MCU (STM32 Blue Pill):** Selected for its 72MHz ARM Cortex-M3 core, which provides the necessary real-time processing speed to parse high-frequency sensor data and output rapid PWM signals.
- **Secondary Processor (Raspberry Pi):** Implemented to run a custom OpenCV pipeline and process camera vision algorithms. The Raspberry Pi handles high-level lane-heading calculations and sends directional commands to the STM32 using serial communication.

Sensors and Navigation

- **Time-of-Flight Sensors (VL53L0X):** Chosen over ultrasonic sensors (HC-SR04) for their laser-based precision, millimeter-level accuracy, and faster response times, which are critical for avoiding obstacles at high speeds.
- **Vision and Lane-Tracking (Wide-Angled Camera Module):** Interfaced directly with the Raspberry Pi to feed real-time visual data into the OpenCV navigation stack. A wide-angled camera lens was selected to ensure that inner and outer track boundaries remain fully visible in the frame, completely mitigating blind spots experienced by the narrow ToF sensors on wide straightaways.
- **IMU (MPU6050):** A 6-axis gyroscope and accelerometer used to track the vehicle's angular velocity. The data provided allows the MCU to detect oscillation

and apply counter-steering logic to help stabilize the chassis during aggressive steering.

- **Boundary and Line Detection (QTR-8RC Reflectance Sensor Array):** Mounted downwards at the front of the chassis to detect track lines and surface boundaries with very low latency. While the Raspberry Pi and camera provide long-range predictive navigation, the QTR-8RC provides instant, high-frequency, short-range data to the STM32. This allows the MCU to execute high-speed PID corrections without waiting for the heavier image processing overhead. The RC variant was selected because its discharge time can be measured using the digital I/O pins without allowing the STM32's hardware ADC channels to be used for other telemetry.

Power Distribution and Motor Drive

- **Power Supply (Dual 18650 Li-ion Cells):** Provides 7.4V, balancing optimal energy density with sufficient mass to lower the vehicle's center of gravity and prevent nose-diving under high-braking conditions.
- **Voltage Regulation (LM2596 Buck Converter):** Steps down the raw 7.4V from the power supply to a stable 5V line to safely power the Raspberry Pi, STM32, and logic sensors without risking thermal throttling.
- **Motor Driver (TB6612FNG):** Selected over the legacy L298N due to its MOSFET-based H-bridges. It operates with higher efficiency, generates minimal waste heat, and has a smaller physical footprint, creating space on the perfboard.
- **Actuation and Odometry (6V GA12-N20 1500 RPM Gear Motors with Encoders):** Selected to achieve a highly competitive speed. Because driving at high speeds increases the risk of traction loss, the integrated encoders are important for closed-loop PID control. They provide real-time wheel RPM feedback to the STM32, ensuring that the wheels remain perfectly synchronized. This prevents spin-outs and maintains precise heading accuracy during high-speed actions.

Safety Implementation

- **Power Cutoff System:** In accordance with the universal design constraints, the main battery power is routed through a primary side-mounted rocker switch and a top-mounted Emergency Push Button (E-Stop) connected in series. This is configured so that standard power is controlled by the rocker switch, but the E-Stop button can instantly override it, physically cutting off power to the circuit when necessary.